

A pre-feasibility study for the establishment of Private school in Jarash.



2017



List of contents

.1 EXECUTIVE SUMMARY	5
KEY HIGHLIGHT OF JORDAN	6
JORDAN INVESTMENT COMMISSION AIMS TO:.....	11
2. MARKET ANALYSIS	12
Project description	12
Project objectives	12
Proposed service	12
Target segments.....	13
Analysis of market size	13
Competitors analysis	13
Price analysis	14
Market share	15
Projected revenues	16
3. TECHNICAL ANALYSIS	17
Human resources required	17
The following that, job descriptions for project:.....	18
Side analysis	19
Technical requirements analysis.....	19
Land and construction works	19
Machinery and equipment	20
Furnitures	20
.4 FINANCIAL ANALYSIS	21
Assumptions.....	22
Capital Expenditures	24
Operating Expenses	27
General and Administrative Expenses	28
Marketing expenses	28
Human Resources	29
Depreciations	31
Loan	32
Income Statement	33
Expected Cashflows Statement	35
Expected Balance Sheet.....	37
Feasibility Indicators	39
Sensitivity Analysis	40

List of tables

Table1 economic indicators.....	5
Table2 Population Distribution according to Age Group:	7
Table3 Income Tax and Exemptions.....	11
Table4 the average instalments at private schools in Jerash.....	14
Table5 Annual instalments rates expected in (JD).....	14
Table6 the school's annual occupancy	15
Table7 Revenues of membership bus (JD)	16
Table8 Projected annual revenue for the school (JD).....	16
Table9 Manpower	17
Table 10 the land required for the project.....	19
Table11 construction required.....	19
Table12 machinery and equipment	20
Table 13 General Assumptions	22
Table 14 Currency Exchange Rates	22
Table 15 Annual Growth Rates Assumptions.....	22
Table 16 Expenses Assumptions.....	22
Table 17 Income Tax Assumptions	23
Table 18 Weighted Average Cost of Capital.....	23
Table 19 Land Capital Expenditure	24
Table 20 Construction Work Capital Expenditure	24
Table 21 machinery and equipments of Capital Expenditure	24
Table 22 furniture of Capital Expenditure.....	25
Table 23 transportation of Capital Expenditure	25
Table 24 Pre Operating Expenses.....	25
Table 25 Working Capital	25
Table 26 Expenses Summary	26
Table 27 Sources of Funding.....	26
Table 28 Operating Expenses	27
Table 29 General and Administrative Expenses	28
Table 30 Marketing Expenses.....	28
Table 31 expected staff numbers and wages (JD)	29
Table 32 Expected Annual Salaries and Wges (2027-2018).....	30
Table 33 Capex and Depreciation Expenses.....	31

List Figures

Figure 1 Population of Jordan	6
Figure2 Sectors Contribution to Jordan's Economy	8

1. Executive summary

The idea of the project is to create an integrated private school in Jerash governorate, to serve the educational sector in the governorate and work to raise awareness for the children of the region, which would be inclusive of all levels of education from kindergarten to high school.

Based on the proposed market share and prices approved by the proposed project, the expected revenues amounted to JD 879,950 during the first year of the project and gradually increases to JD 1,287,791 during the tenth year. The period of redemption of the capital is five years with an internal rate of return of 20.66. In addition to providing 66 job opportunities for the people of the governorate.

The following table shows the economic indicators of the project:

Table1 economic indicators

Indicator	Value
Total Investment Cost (Jordan Dinars)	1,910,807
Internal Rate of Return (IRR)	%20.66
Pay-Back Period (Years)	5
Net Present Value (Jordan Dinars)	182,437
Job Opportunities provided by the Project	66
Total Revenues (For 10 Years)	11,021,530
Total Operating Expenses (For 10 years)	3,929,931
Total Net Profit (For 10 years)	2,734,584

Key Highlight of Jordan

Jordan Hashemite Kingdom area is 89342 Kilometers with a population exceed 9.8 Millions distributed among 12 Governorates.

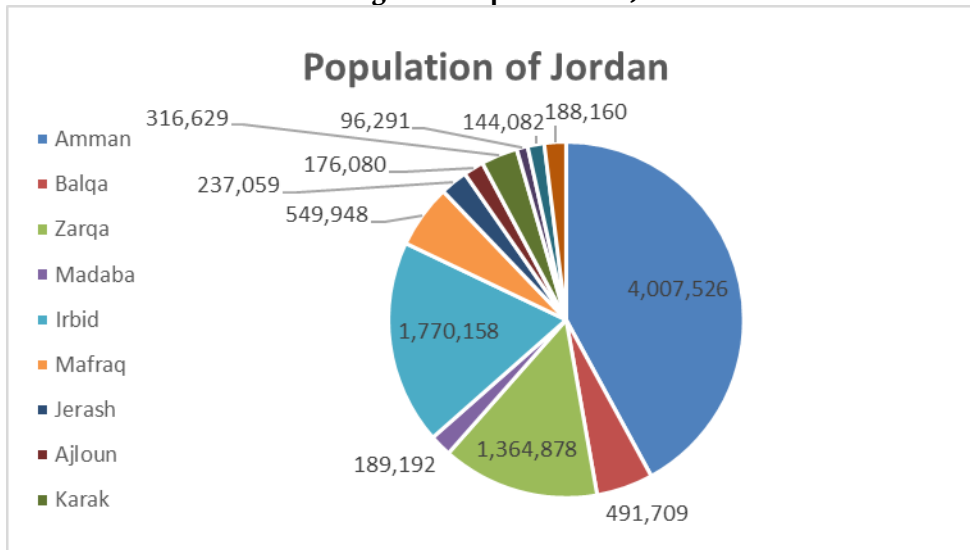
Jordan is bounded by Saudi Arabia, Iraq, Syria and Palestine from the South, East, North and west respectively. Jordan climate is mostly arid desert with rainy season between November to April Months.

Jordan is a free market economy, ranked as the fifth freest economy in the MENA region in the Index of Economic Freedom. Its free market economy enjoys strong partnerships amongst its neighboring country as well as Europe and the USA. Jordan is a signatory of several bilateral and multilateral trade agreements such as (Free Trade Agreement with the US and the EU, a Free Trade Agreement with the EFTA states, an FTA with Singapore, A member of the Greater Arab Free Trade Agreement (GAFTA) and a signatory of the Aghadir Agreement. Looking at the range of countries these agreements cover and facilitate trade between, one is confident to say that Jordan has business gateways and partnerships across the globe.

Population Overview

Based on the latest surveys done by Department of Statistics (DOS) in 2017, total number of population has reached around 9.814.995 with an increase rate of 2.88% from 2016 year, figure below summarizes the population distribution among the kingdom governorates.

Figure 1 Population of Jordan



Department of Statistics (DOS), 2015

Moreover, and based on the latest survey conducted by (DOS) which related to population nationality distribution over the governorates, it was found that total Non-Jordanian residents constitutes around 30% of the total population. Half of these are Syrians (1.3 Million) concentrated mainly in Amman Capital (436 Thousand) then Irbid (343 Thousand), Mafraq at 208 Thousand and Zarqa at 175 Thousand.

Jordan prides itself in its youthful population with 35% of ages below 15 years old and only 4% with ages above 65 years old. Table below summarizes population distribution of Jordanians according to Age Group:

Table2 Population Distribution according to Age Group:

Age Group	Population %	Males	Females
0-14 Years	%35.04	%19.2	%18.2
15-64 Years	%61.02	%30.7	%28.6
More than 65 Years	%3.94	%1.6	%1.6

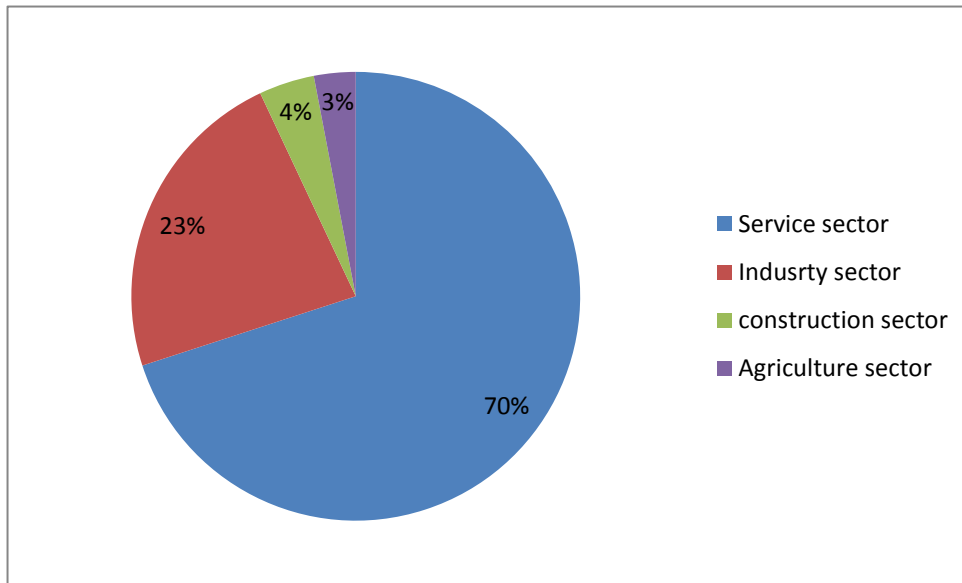
Department of Statistics (DOS), 2015.

Overview of Jordan Economy

Classified as an upper middle economy by the World Bank and as a country of High Human Development by the UNDP, Jordan is an important financial power in the Middle East. Its small market witnessed growth in the last two decades since King Abdullah II accessed the throne. Jordan's GNI per capita has increased in the last four years due to the basket of economic reforms that were enacted in the recent decade such as (the new Income Tax Law, Public Private Partnership Law and Investment Law). These laws aim to encourage the participation of the private sector in the Kingdom's economic development and provide an enabling legislative environment for current and new investment opportunities.

The Jordanian economy is overwhelmingly services oriented and its contribution in the GDP is 68.1%. The contribution of the industrial sector is 22.4%, the construction sector is 4.2% and the agricultural sector is 2.9%. These contributions are aimed to increase to (27.4%), (5.8%) and (3.4%) respectively and that of the service sector is due to decrease according to Jordan 2025 vision.

Figure2 Sectors Contribution to Jordan's Economy



Jordan's exports include variety of textiles, potassium, phosphates, fertilizers, vegetables and pharmaceutical products. While it's main imports are crude and refined petroleum. Distinguished by its strategic location, on the crossroads between Asia, Africa and Europe with strong connections to the Levant and the GCC, Jordan has a regional market of interest that represents US\$3.8 trillion market and compromising 380 million consumers.

Jordan infrastructure ranks comparatively well (38th out of 148 comparable economies), with an extensive 8000 KM road network connecting Jordan domestically and externally The new Queen Alia International Airport and the Port of Aqaba are the major gateways to the international market. In addition to some mega projects such as the Red- Dead Sea Canal and the national railway network that will be developed to position Jordan as a hub for regional commerce.

Jordan banking system is quite sophisticated, resilient and in compliance with international standards, making it very attractive and trustworthy to investors. This is reflected in the fact that 50% of equity in licensed banks in Jordan is held by non-Jordanians, and non residents' deposits in Jordanian banks witnessed a steady growth of 19.2%.

Moreover, realizing the value of MSMEs to drive economic growth, the government has developed the national Strategy for the encouragement of entrepreneurship and the development of micro small and

medium sized enterprises for 2015-2019. This shows the commitment of the Jordan government to enhance the private sector development and leveraging the country's strong human capital.

Jordan prides itself in its youthful population. It's the country most valuable capital. A tech savvy well educated and trained workforce attracts a lot of investors to Jordan. More than 20.4% of Jordan's GDP is dedicated to the education and capacity building for the labor force, this has resulted in securing a 91% literacy rate and enabled Jordanians to be among most hired and qualified middle-eastern workforce.

Such solid, diverse and resilient characteristics of the Jordanian economy and investment scene position Jordan as a competitive investment destination.

Major Economic Indicators

- **GDP Growth**

The Jordanian economy slowed and reached 2.4 percent in 2015 compared to growth rate of 3.1 in 2014 and similar to MENA growth rates.

The growth also slowed for trade, restaurants and hotels; manufacturing; transport; and electricity and water. Meanwhile, finance, insurance, and business services advanced at a faster pace.

- **Inflation rate**

Inflation rate decreased to 0.9 in 2015 after 2% decline in the previous year. Inflation Rate in Jordan averaged 3.1 percent from 2011 until 2015.

- **Unemployment**

The unemployment rate in Jordan was recorded at 13 percent in 2015, comparing to 11.9 percent a year earlier. This increase due to the current instability in the labor market structure as well as high competition from low cost foreign labors especially from Syria.

- **External Sector**

A number of risks manifested in 2015, the closure of land trade routes with Syria and Iraq and the deepening instability in the region adversely impacted many external sector indicators such as trade, tourism and direct Investment.

Moreover, loans disbursements increased by JD 545.3 million, due to the use of the International and Arab Monetary Funds (IMF and AMF) credit facilities. As an outcome of these developments, the overall balance of the balance of payments registered a surplus of JD 328.7 million in 2015, compared with a surplus of JD 1,550.7 million in 2014.

Further, net international investment position (IIP) witnessed an increase in the Kingdom's net obligations to abroad; to reach JD 24,357.5 million, compared to a net obligation of JD 22,578.8 million at the end of 2014 as a result of the increase in the stock of external financial assets and liabilities of all resident economic sectors to reach JD 18,657.9 million and JD 43,015.5 million; respectively.

External Debt in Jordan increased to 9390.50 JOD Million in 2015 from 8030.10 JOD Million in 2014. External Debt in Jordan averaged 5433.67 JOD Million from 1988 until 2015, reaching an all time high of 9390.50 JOD Million in 2015 and a record low of 3640.20 JOD Million in 2008.

Investment Climate in Jordan

Investment in Jordan is considered as one of the vital sources to boost the economy considering that Jordan's economy is among the smallest in the Middle East, with insufficient supplies of water, oil, and other natural resources, underlying the government's heavy reliance on foreign assistance.

The country's location, supported by myriad free trade agreements (FTAs) offering access to 1.5bn consumers, enables the kingdom to be a strategic trade route to many of its neighboring countries and regions.

Jordan aspires to create a competitive investment destination capitalizing on its many advantages mentioned above. The government focused its efforts to implement significant advances in structural and legal reform. These efforts are represented in the new Investment Law of 2014, the tax law, in addition to other endeavors related to providing greater access to credit for MSMEs, and by providing greater investment incentives to specific priority sectors identified by the new Investment law and the Jordan 2025 vision.

furthermore, in it endeavor to enhance the business environment, several geographically industrial states were created across the kingdom. These range in type to include (industrial estates, free zones and special economic zones). Under the new Investment Law, these zones are given a number of incentives that include a tax rate of 0% on exports, sales tax, import duties, social service tax, dividends tax and a 5% income tax from all economic and manufacturing activities undertaken in the development zones. As for the investments in Free Zones, they enjoy Exemptions from customs duties, income tax exemptions and exemptions from land and building taxes among others. Both development and free zones also enjoy facilitations regarding visa and residency permits for investors and workers in addition to Repatriation of capital and profits in a convertible currency. Such estates have succeeded in attracting relatively large amounts of FDI to Jordan.

Key National Investment Priorities:

Jordan Investment commission worked on taking a leading role in the application of government policies to promote and attract domestic and foreign investments and create an investment environment that stimulates economic performance through investment promotion strategy launched in 2016

Jordan investment commission aims to:

- Regulating the special provisions governing Development Zones and Free Zones in the Kingdom and developing them placing them in service of the national economy as well as monitoring their functioning.
- Developing plan and programs to stimulate domestic and foreign investments.
- Establishing trade centers and organizing exhibitions as well as opening markets and organizing trade missions in order to promote national products, in addition to marketing and development of national exports and encouraging investment.
- Taking appropriate decisions related to private or public institutions to improve Investors' confidence in Jordan's investment environment.

Investment Climate in Jarash:

Under the provisions of fifth article of the investment law No. 30 of 2014 and Income tax regulations in the less developed areas, the reduction of the income tax system on the investment projects are clarified in the table below:

Table3 Income Tax and Exemptions

Within the Development Zone or Free Zone	Less Developed Areas
Income Tax is 5%	Income Tax is 20% As per Investment Law categorization, Irbid is classified in Class (C) where investors enjoys 60% exemption on Income tax for 20 years, where net income tax after reduction should not be less than 5%.

2. Market analysis

Project description

The project includes the establishment of a private high school in the governorate of Jerash because of the increasing numbers of students and demand for private schools in the governorate, especially as it lacks a private schools that include 11th and 12th grades.

This school will be unique in both services and education. It will include highly qualified teachers with a high level of experience in the teaching sector. It will include kindergarten, primary and secondary level. The total area of the project is 8,000 sq.m containing classrooms, administration, warehouses, library, a music room, computer and science labs, two indoor and outdoor courts and a clinic, in addition to playgrounds.

Project objectives

The proposed project aims to:

- Provide high level of educational services for the governorate.
- Increase the educational level and cultural awareness of the children of the governorate.
- Creation of job opportunities for the people of the governorate and improve living and social standards for the workers in the project.
- Encouraging investment in the educational sector and achieving a good return for the project owner.

Proposed service

The proposed project will establish a private school in Jerash governorate, which will be equipped with all educational services, equipment and modern technology to be a unique school within the governorate.

The school will include all stages of education. It will include the kindergarten stage, which is from 4 years to 6 years. The primary stage, which is from the age of 6 years to the age of sixteen (i.e from the first to the tenth grade), in addition to the secondary stage with scientific and literary specialties.

Target segments

The project targets the children of the governorate in general and especially students aged between 15 and 18 years because the governorate lacks a strong secondary male school in the terms of education.

Analysis of market size

The population of Jerash governorate is 237,059, according to the latest study conducted by the Department of Statistics in 2015. The number of students is about 48,000, of whom 5,000 are in private schools.

School dropout rate in Jerash governorate was (0.49%), which is higher than the Kingdom (0.25%).

The rate of success in secondary school in Jerash governorate was (35.6%), which is lower than the Kingdom (36.6%).

The percentage of illiteracy in Jerash governorate was 8.8%, which is higher than the Kingdom (7%).

The student / teacher rate index in Jerash governorate was 13.64, which is lower than the Kingdom, estimated at 16.12 students.

All of the indicators above indicate that there is a need to establish additional educational projects in the governorate.

Compititors analysis

The total number of schools in Jerash governorate is 173, of which 47 are rented schools. The number of students is about 48,000, of whom 5,000 are students in private schools. The number of teachers in the governorate is 3152 teachers.

There are 53 schools in Jerash governorate. The Jerash Secondary Private School is one of the best schools in the governorate. The number of its students is about 382 students and includes the first grade to the tenth grade.

There are also many secondary schools that include industrial and vocational specialties, including:

- Hasan Al - Kayed Vocational Secondary School for Boys.
- Blilla Secondary School for Boys..
- Blilla High School for Girls.
- Al - Qabsi Secondary School for Boys.
- Mersa' Secondary School for Boys.
- Mersa' Secondary School for Girls..

Price analysis

Pricing policy is an important element of marketing, which determines the project's ability to attract target segments. According to the study of private schools' prices in the governorate and in proportion to the services provided, it was found that the prices of premiums in private schools in the governorate are as follows:

Table4 the average instalments at private schools in Jerash

Educational level	Annuity value (JD)
Kindergarten 1- Kindergarten 2	300 -200
First grade- Seventh grade	525 -350
Eighth grade- tenth grade	650-550

Accordingly, the proposed prices for the project were set as follows:

- The rate of subscription fees for buses was assumed to be 350 JD per year for all classes and distances.
- The projected premiums of the school at the start of its operation, the following table shows prices for each grade:

Table5 Annual instalments rates expected in (JD)

Grade	Annuity (JD)
Kindergarten 1	300
Kindergarten 2	375
First grade	450
Seconde grade	475
Third grade	500
Fourth grade	550
Fifth grade	575
Sixth grade	600
Seventh grade	625
Eigth grade	675
Ninth grade	700
Tenth grade	750
Eleventh grade	800
Twelfth grade	900

Market share

Through the study of the project and comparison with other schools, it was assumed that the capacity of one grade with different classrooms is about 82 students, equivalent to 7% of the total capacity of the school and this percentage represents the kindergarten to the eighth grade, Where the the rate of enrollment will be risen For one class with its classrooms is about 88 students, equivalent to 7.5% of the total capacity of the school ,this ratio increase represents the ninth grade to the twelfth grade, The reason of the increase is due to the need of the governorate to a private high school with an experienced staff using good teaching methods.

The occupancy rates during the first ten years of the project are as follows:

Table6 the school's annual occupancy

Occupancy rate and number of total students	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Ratio of students of full capacity	65%	70%	75%	75%	80%	85%	85%	90%	90%	95%
Number of students	1170	1260	1350	1350	1440	1530	1530	1620	1620	1710

Projected revenues

.Projected annual revenues for the project

The school's revenues are mainly based on student fees and bus subscriptions, as well as other income such as rental of the canteen.

Annual income is the total number of students per class multiplied by the amount of the premium allocated, taking into account the annual increase in premiums and transportation by 3% for the following year.

For bus subscription revenues, it is assumed that the number of students subscribed are 40% of the total number of students, therefore the value of the revenues from the bus subscription is the total number of students in the school multiplied by 40% multiplied by the value of the bus subscription. The following table shows the school revenues from bus subscriptions:

Table7 Revenues of membership bus (JD)

Bus service	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Price	350	360.5	371	382	394	406	418	430	443	457
Number of students of membership	468	504	540	540	576	612	612	648	648	684
Bus service revenues	163800	181692	200510	206525	226903	248317	255766	278935	287303	312363

The following table summarizes the projected annual revenues of the school for ten years:

Table8 Projected annual revenue for the school (JD)

Price assumptions	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Instalments revenues	696150	772191	827348	827348	882504	937661	937661	992817	992817	1047974
Membership bus revenues	163800	168714	173775	178988.68	184358	189889	195586	201453	207497	213722
Side revenues	20000	20600	21218	21855	22510	23185	23881	24597	25335	26095
Total	879950	961505	1022341	1028191	1089373	1150735	1157127	1218868	1225649	1287791



3. Technical Analysis

This section aims to clarify the technical specifications of the project in addition to estimating and reviewing the most important parts of investment and operating costs based on the results of the market study.

The technical analysis includes a general description of the aspects of the project and its components, as well as the calculation of the construction and operational requirements of the project, providing the general specifications on which the costs were based, according to the technical team.

Human resources required

This project requires a general manager to supervise all educational matters, in addition to the required educational and administrative staff, including teachers and social counselors, a financial manager and an accountant who manages the school's finances in addition to a doctor.

The following table shows the required staff for the project:

Table9 Manpower

Job title	No.
School manager	1
Secretary	2
Financial manager	1
accountant	2
Doctro	1
Driver	9
Security	3
Cleaners	10
Total employment of indirect	25
Teacher	35
Social worker	2
Total employment of direct	37
Total	66

The following is the job descriptions for the staff of the project

- **School manager:** Direct supervision of the educational process and ensure the health of students during the school day and the implementation of the curriculum, follow up the absences and delay of staff, and meet parents in the necessary needs.
- **Secretary:** receive the school correspondence and transactions and presente it to the school manager and take the necessary procedures in accordance with the guidance of the director.
- **Financial manager:** Identifying organizational objectives, planning, organizing and implement financial plans according to the rules and procedures.
- **Teacher:** Respecting the student and treat him in an educational manner to develop his personality and self –confidence. Teaching and do whatever is required to achieve the objectives of articles that examine of preparing and teaching methods and the methods of evaluation, testing, and activities inside and outside the classroom, as required by the base of the profession and the course, in accordance with regulations and directives received from competent authorities.
- **Social counselor:** work on achieving the objectives of education in regard to student affairs and registration of new students and follow up with them, keeping students files in their places and archiving them, in addition to preparing the attendance record and the daily absence of all students and inquire about the reasons for the absence.
- **Accountant:** Responsible for the calculation of school costs in addition to record all financial transactions
- **Doctor:** treat students and provide health care In case of any accident.
- **Drivers:** picking up students from home to school, and vice versa, as well as ensuring the safety of students through the safety of driving.
- **Security:** will maintain the security of the school.
- **Cleaner:** will maintain the cleanliness of classrooms, offices and playground.

Side analysis

Based on market analysis, two suitable locations for the project were proposed:

- Deir Al-Liat, located in the north-west of Jerash
- Green Mountain area, which is considered one of the finest areas in the governorate in addition to the area of Shawad, which is located near the governorate building.

Based on this, it is proposed that the project will be implemented in Shawahed area because of the moderation of the land prices. In addition, it is an area serviced by all necessary services, besides being close to Jerash governorate building.

The average price per square meter is between (35-100) JD.

Technical requirements analysis

Land and construction works

The project will be constructed on an area of 8000 sq.m. The following table shows the needed land.

Table 10 the land required for the project

Land	Area(m2)
Land required	8,000

The following table shows the required construction work for the project:

Table11 construction required

structural works	Area (m2)
Constructions works	3,000
Building design	3,000
Excavation work	3,000
Equipping playgrounds and parks	2,500

Machinery and equipment

The following table shows the required machinery and equipment needed for the project:

Table12 machinery and equipment

Machines and equipment	No.
Computers	50
Central air condition	1
Labs	2
clinic	1

Furnitures

The estimated furniture based on the number of classes and total number of students in the school, along with the required administrative and educational staff.

4. Financial Analysis

The Financial Part illustrates all the assumptions which were used when we have developed the study including project expenses related to capital expenditures, operating expenses (Fixed and Variable costs) followed by illustrating project revenues.

Assumptions

Calculation Assumptions:

- The weighted average cost of capital used in the study is 14.81% as it was calculated considering risk free and debt Interest rates as well as Market risk premiums to consider alternative opportunity costs for investor.
- Income tax on the net project income is 20 % during the life time of the project.
- Project working capital has been estimated and added to project cash outflows at the beginning of the project and annual increase in working capital has been also estimated and added to the operating expenses then net working capital has been added to total cash inflows at the end of project lifetime.

The time span for the financial study is 10 years starting from 2018, where annual increase rates have been estimated as per the tables below:

Assumptions

Table 13 General Assumptions

General Information	
All Financial Numbers (Currency)	Jordan Dinars
Financial Study Time Span (Years)	10
Expected Project First Operating Year	2018
Last Year in the Financial Study	2027

Table 14 Currency Exchange Rates

Exchange Rates	Jordan Dinars
American Dollar	0.708
Euro	0.760

Table 15 Annual Growth Rates Assumptions

Annual Growth Rates	Average
Annual Population Growth Rate	2.20%
Annual Sales Price Increase Rate	3.00%
Annual Expenses Increase Rate	3.00%
Annual Increase Rate in Employees Salaries	4.1%

(1) and (2) Source: Inflation Rates, Central Bank of Jordan

(3) Source: Social Security

Table 16 Expenses Assumptions

Expenses Assumptions	
Utilities Expenses from Total Revenues	5%
Maintenance Expenses from Total Revenues	0.5%
Depreciation Expenses from Total Revenues	1%
Insurance Expenses from Total Assets Value	0.5%
Marketing Expenses from Total Revenues	0.5%

Table 17 Income Tax Assumptions

Income Tax Assumptions	Value
(1) Average Income Tax in Jordan	%20
(2) Income Tax Deduction	%0
Income Tax after Deduction	%20
(3) Compulsary Reserve Percentage	%10
Other Assumptions	Value
(1) Annual Monthly Salaries have been calculated after multiplying monthly salaries by:	14

Risk Premuim	
(1) Risk Free Rate of Return	6.50%
(2) Return to Debt Maturity	9.48%
(3) Market Risk Premuim	10.12%
Income Tax Rate	20%
(4) Beta	1
Growth Rate	4.00%

(1) Source :Damodaran's Country Default Spreads and Risk Premiums Report

(2) Source :Central Bank of Jordan

(3) Source :Damodaran's Country Default Spreads and Risk Premiums Report

(4) Source :Damodoran Beta By Sector, , pages.stern.nyu.edu

Table 18 Weighted Average Cost of Capital

Weighted Average Cost of Capital	
Average Return on Risk Free Investment	%6.50
Return to Debt Maturity	%9.48
Market Risk Premuim	%10.12
Income Tax Rate	%20
Bets	1
Equity	1,676,021
Loans Value	419,005
Loans	
Cost of Borrowing before Tax	%9.48
Cost of Borrowing After Tax	%7.58
Loans Value	419,005
Loans Percentage	%20

Equity	
Cost of Equity	%16.6
Equity Value	1,676,021
Equity Percentage	%80
Gross	
Project Value	2,095,026
Weighted Average Cost of Capital	%14.81

Capital Expenditures

The estimated cost of the project is 2,095,026 JD and it covers land costs, construction works, machinery and equipment, furniture, pre-operating expenses and working capital. The following tables summarize the details of these expenses:

Table 19 Land Capital Expenditure

Land	Cost Jordan Dinar/(Meters Square)	Area (M2)	Cost
Required Land	50	8,000	400,000
Cost			400,000

Table 20 Construction Work Capital Expenditure

Description	Cost (JOD per Squared)	Area	Gross Cost
Construction Works	250	3,000	750,000
Design and blueprints	7	3,000	21,000
Excavation	5	3,000	15,000
Equipping playgrounds and parks	30	2,500	75,000
Other Works			11,100
Cost			872,100

Table 21 machinery and equipments of Capital Expenditure

Description	No.	Cost (JOD per unit)	Gross cost
Computers	50	300	15,000
Central air conditioner	1	25,000	25,000
Labs	2	15,000	30,000
Clinic	1	3,500	3,500
Cost			73,500

Based on the market survey, the average cost of chairs per student costs 50 dinars after taking into account the existence of Office furniture and educational staff and others. The following table shows the total value of furniture for school:

Table 22 furniture of Capital Expenditure

Description	No.	Cost(JOD per unit)	Gross cost
Furniture	-	-	60,000
Total			60,000

Customize to school buses transport to students and by rate is suitable to the total number of students. The following table shows the number and cost of transport to the project

Table 23 transportation of Capital Expenditure

Description	No.	Cost(per unit)	Gross cost
Buses transport	9	40,000	360,000
Cost			360,000

Table 24 Pre Operating Expenses

Pre-operating expenses	Estimated cost (JD)
Government fees	1,500
Transportation	1,500
Marketing	1,500
Training	1,500
Total	6,000

The initial working capital reflects the project's amounts needed to cover all the expenses during operation until the project starts generating income, the following table shows the components of the initial working capital

Table 25 Working Capital

Working Capital	No. of Months	%	Estimated Cost
Operating expenses	3	%25	80,849
General and administrative expenses	3	%25	9,820
Marketing expenses	3	%25	1,100
Indirect Salaries	3	%25	41,200
Total			132,970

Table 26 Expenses Summary

Capital expenditure summary	Cost (JD)	Contingency	Cost (JD)	%
Land	400,000	10%	440,000	%21.0
Construction works	872,100	10%	959,310	%45.8
Machinery & Equipment	73,500	10%	80,850	%3.9
Furniture	60,000	10%	66,000	%3.2
Vehicles	360,000	10%	396,000	%18.9
Pre-Operating Expenses	6,000	10%	6,600	%0.3
Working capital	132,970	10%	146,266	%7.0
Total (JD)	1,904,570		2,095,026	100%

Sources of Funding

The financing structure of the project consists of loans and Equity. The investment cost of this project is estimated at (2,095,026) JD, where 20% of the project will be financed through loans (419,005) JD, and the remaining (80%) through (Equity) with (1,676,021) JD. The following table summarizes the general structure of the required financing:

Table 27 Sources of Funding

Funding Sources	Amount	Percentage
Equity	1,676,021	%80
Loans	419,005	%20
Total	2,095,026	%100
Use of Fund:	Value	Ratio
Capital Expenditures	1,942,160	%92.70
Pre-Operating Expenses	6,600	%0.32
Working Capital	146,266	%6.98
Total	2,095,026	%100.00

Operating Expenses

The following table summarizes the expected operating expenses for the proposed project (2018-2027):

Table 28 Operating Expenses

Description	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Raw Materials and Packaging from Revenues	0	0	0	0	0	0	0	0	0	0
Utilities from Revenues	43,998	48,075	51,117	51,410	54,469	57,537	57,856	60,943	61,282	64,390
Maintenance from Revenues	4,400	4,808	5,112	5,141	5,447	5,754	5,786	6,094	6,128	6,439
Disposables from Revenues	8,800	9,615	10,223	10,282	10,894	11,507	11,571	12,189	12,256	12,878
Direct Salaries	236,800	246,509	256,616	267,137	278,090	289,491	301,360	313,716	326,578	339,968
Others	29,400	30,901	32,307	33,397	34,890	36,429	37,657	39,294	40,625	42,367
Total	323,396	339,907	355,375	367,366	383,789	400,718	414,231	432,237	446,870	466,042

General and Administrative Expenses

The following table summarizes the expected general and administrative expenses for the proposed project (2018-2027):

Table 29 General and Administrative Expenses

Description	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Telecom	2,400	2,472	2,546	2,623	2,701	2,782	2,866	2,952	3,040	3,131
Hospitality	1,800	1,854	1,910	1,967	2,026	2,087	2,149	2,214	2,280	2,349
Stationary and Printing	3,600	3,708	3,819	3,934	4,052	4,173	4,299	4,428	4,560	4,697
Transportation	17,200	17,716	18,247	18,795	19,359	19,940	20,538	21,154	21,788	22,442
Training	1,000	1,030	1,061	1,093	1,126	1,159	1,194	1,230	1,267	1,305
Insurance	9,711	10,002	10,302	10,611	10,930	11,257	11,595	11,943	12,301	12,670
Miscellaneous	3,571	3,678	3,789	3,902	4,019	4,140	4,264	4,392	4,524	4,659
Total	39,282	40,460	41,674	42,924	44,212	45,538	46,905	48,312	49,761	51,254

Marketing expenses

The following table shows the expected marketing expenses of the proposed project. Marketing expenses account for 0.01% of revenues based on the team's estimate:

Table 30 Marketing Expenses

Description	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Total	4,400	4,808	5,112	5,141	5,447	5,754	5,786	6,094	6,128	6,439

Human Resource

The total annual salaries and monthly wages has been estimated on 16 months for the purpose of considering annual increments on salaries as well as payments related to social security and health insurance,... etc.

The following table summerizes the expected annual salaries and wages, considering annual increment on salaries as per the income statement:

Table 31 expected staff numbers and wages (JD)

Job Title	No.	Monthly Salary	Sum of monthly	Sum of yearly salaries
General manager	1	1,500	1,500	24,000
secertaria	2	350	700	11,200
Financial manager	1	750	750	12,000
Accountant	2	400	800	12,800
Doctor	1	600	600	9,600
Driver	9	300	2,700	43,200
Sequiry	3	250	750	12,000
Cleaning labours	10	250	2,500	40,000
Indirect employes	29		10,300	164,800
Teachers	35	400	14,000	224,000
Social officer	2	400	800	12,800
Direct employes	37		14,800	236,800
Total	66		25,100	401,600

The following table shows the expected annual salaries for employees for 10 years of the life of the project:

Table 32 Expected Annual Salaries and Wges (2027-2018)

Job title	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
موظفين غير مباشرين										
General manager	24,000	24,984	26,008	27,075	28,185	29,340	30,543	31,796	33,099	34,456
secertaria	11,200	11,659	12,137	12,635	13,153	13,692	14,254	14,838	15,446	16,080
Financial manager	12,000	12,492	13,004	13,537	14,092	14,670	15,272	15,898	16,550	17,228
Accountant	12,800	13,325	13,871	14,440	15,032	15,648	16,290	16,958	17,653	18,377
Doctor	9,600	9,994	10,403	10,830	11,274	11,736	12,217	12,718	13,240	13,782
Driver	43,200	44,971	46,815	48,734	50,733	52,813	54,978	57,232	59,579	62,021
Sequiry	12,000	12,492	13,004	13,537	14,092	14,670	15,272	15,898	16,550	17,228
Cleaning labours	40,000	41,640	43,347	45,124	46,975	48,901	50,905	52,993	55,165	57,427
Indirect employes	164,800	171,557	178,591	185,913	193,535	201,470	209,730	218,329	227,281	236,599
Teachers	224,000	233,184	242,745	252,697	263,058	273,843	285,071	296,758	308,926	321,592
Social officer	12,800	13,325	13,871	14,440	15,032	15,648	16,290	16,958	17,653	18,377
Direct employes	236,800	246,509	256,616	267,137	278,090	289,491	301,360	313,716	326,578	339,968
Total	401,600	418,066	435,206	453,050	471,625	490,961	511,091	532,046	553,859	576,568

Depreciations

The following tables illustrate cost of Capex and Depreciation rates:

Table 33 Capex and Depreciation Expenses

Capex Cost and Annual Depreciation Rates	Cost (JOD)	Depreciation Rate	Annual Additions Percentage
Land	440,000	%0.0	%0.0
Construction Works	959,310	%5.0	%0.0
Machinaries	80,850	%10.0	%2.0
Furniture	66,000	%10.0	%5.0
Vehicles	396,000	%10.0	%0.0

Table 1 Depreciations and Additions on Construction Works

Capex, Annual Additions and Depreciations	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Total Fixed Assets	1,942,160	1,947,077	1,952,191	1,957,512	1,963,048	1,968,810	1,974,807	1,981,050	1,987,551	1,994,321
Total Depreciation Expenses	102,251	102,742	103,254	103,786	104,339	104,915	105,515	106,139	106,790	107,467
Total Accumulated Depreciation	102,251	204,993	308,246	412,032	516,371	621,287	726,802	832,941	939,731	1,047,198
Total Additions	0	4,917	5,114	5,321	5,536	5,761	5,997	6,243	6,501	6,770
Total Net Book Values	1,839,910	1,742,084	1,643,945	1,545,480	1,446,677	1,347,523	1,248,005	1,148,108	1,047,820	947,123

Loan

The table below summarizes all details related to the loan such as annual installments and payment methods for the remaining amount of the loan for each year of the project.

Table : 2 Loan Details

Loan Value		419,005
Annual Interest Rate		9.48%
Loans Period		5
Loan Starts at year:		2017
Annual Payment		109,068
Number of Payments		5

Year	Annual Payment	Interest	Capital	Loan Remaining Value
2017				419,005
2018	109,068	39,722	69,346	349,659
2019	109,068	33,148	75,920	273,739
2020	109,068	25,950	83,118	190,621
2021	109,068	18,071	90,997	99,624
2022	109,068	9,444	99,624	0

Income Statement

All financial information and analysis of the financial expenses of the project shall be the estimated income statement for the years (2018-2027) as shown in the following table:

Table 3 Income Statement

Description	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Revenues										
Sales Revenues	879,950	961,505	1,022,34	1,028,19	1,089,37	1,150,73	1,157,12	1,218,86	1,225,64	1,287,79
Gross Operating Revenues	879,950	961,505	1,022,34	1,028,19	1,089,37	1,150,73	1,157,12	1,218,86	1,225,64	1,287,79
Operating Expenses	323,396	339,907	(355,375)	(367,366)	(383,789)	(400,718)	(414,231)	(432,237)	(446,870)	(466,042)
Gross Operating Profit	556,554	621,598	666,966	660,824	705,584	750,017	742,896	786,631	778,779	821,749
<i>Gross Profit Percentage</i>	<i>%63</i>	<i>%65</i>	<i>%65</i>	<i>%64</i>	<i>%65</i>	<i>%65</i>	<i>%64</i>	<i>%65</i>	<i>%64</i>	<i>%64</i>
Salaries and Benefits (Indirect Staff)	164,800	171,557	(178,591)	(185,913)	(193,535)	(201,470)	(209,730)	(218,329)	(227,281)	(236,599)
General and Administraive Expenses	(39,282)	(40,460)	(41,674)	(42,924)	(44,212)	(45,538)	(46,905)	(48,312)	(49,761)	(51,254)
Markeing Expenses	(4,400)	(4,808)	(5,112)	(5,141)	(5,447)	(5,754)	(5,786)	(6,094)	(6,128)	(6,439)
Pre Operating Expenses	(6,600)									
Gross Indirect Expenses	215,082	216,825	(225,376)	(233,978)	(243,194)	(252,762)	(262,421)	(272,736)	(283,170)	(294,292)
Income before Interest, Depreciation and	341,472	404,773	441,590	426,846	462,390	497,255	480,476	513,896	495,609	527,456
Fixed Assets Depreciations	102,251	102,742)	(103,254)	(103,786)	(104,339)	(104,915)	(105,515)	(106,139)	(106,790)	(107,467)
Income before Tax and Interests	239,221	302,031	338,336	323,061	358,050	392,339	374,960	407,756	388,819	419,990
Bank Interests	(39,722)	(33,148)	(25,950)	(18,071)	(9,444)	(0)	(0)	(0)	(0)	(0)
Income Before Tax	199,500	268,883	312,386	304,990	348,606	392,339	374,960	407,756	388,819	419,990

Description	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Income Tax	(39,900)	(53,777)	(62,477)	(60,998)	(69,721)	(78,468)	(74,992)	(81,551)	(77,764)	(83,998)
Net Profit	159,600	215,107	249,909	243,992	278,885	313,872	299,968	326,205	311,055	335,992
<i>Net Profit Percentage</i>	<i>%18</i>	<i>%22</i>	<i>%24</i>	<i>%24</i>	<i>%26</i>	<i>%27</i>	<i>%26</i>	<i>%27</i>	<i>%25</i>	<i>%26</i>
Compulsory Reserves	(15,960)	(21,511)	(24,991)	(24,399)	(27,888)	(31,387)	(29,997)	(32,620)	(31,106)	(33,599)
Retained Earnings	143,640	337,236	562,153	781,746	1,032,74	1,315,22	1,585,19	1,878,78	2,158,73	2,461,12

Expected Cashflows Statement

The table below summarizes project cashflows statement, where net cash flows are positive during all years as below:

Table 4 Expected Cashflows

	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Cash Inflows from Operating Activities										
Net Profit	159,600	215,107	249,909	243,992	278,885	313,872	299,968	326,205	311,055	335,992
Bank Interests	39,722	33,148	25,950	18,071	9,444	0	0	0	0	0
Depreciation	102,251	102,742	103,254	103,786	104,339	104,915	105,515	106,139	106,790	107,467
Total Operating Cashflows before Additions to Working Capital	301,572	350,996	379,113	365,848	392,668	418,787	405,484	432,344	417,845	443,458
Inventory (Increase/Decrease)	0	0	0	0	0	0	0	0	0	0
Accounts Receivables (Increase/Decrease)	0	0	0	0	0	0	0	0	0	0
Accounts Payables (Increase/Decrease)	53,899	2,752	2,578	1,999	2,737	2,822	2,252	3,001	2,439	3,195
Working Capital (Increase/Decrease)	53,899	2,752	2,578	1,999	2,737	2,822	2,252	3,001	2,439	3,195
Net Cashflows from Operating Activities	355,471	353,748	381,691	367,847	395,406	421,609	407,736	435,345	420,284	446,654
Cashflows from Investments Activities										
Fixed Assets (Procurement)	(1,942,160)	(4,917)	(5,114)	(5,321)	(5,536)	(5,761)	(5,997)	(6,243)	(6,501)	(6,770)

	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Net Cashflows from Investments	(1,942,160)	(4,917)	(5,114)	(5,321)	(5,536)	(5,761)	(5,997)	(6,243)	(6,501)	(6,770)
Activities										
Cashflows from Financing Activities										
Capital	1,676,021									
Loan Amortization	(69,346)	(75,920)	(83,118)	(90,997)	(99,624)	0	0	0	0	0
Bank Interest Rate	(39,722)	(33,148)	(25,950)	(18,071)	(9,444)	(0)	(0)	(0)	(0)	(0)
Loans	419,005	0	0	0	0	0	0	0	0	0
Net Cashflows from Financing Activities	1,985,958	(109,068)	(109,068)	(109,068)	(109,068)	0	0	0	0	0
Net (Increase/Decrease) in Cash	399,270	239,763	267,508	253,458	280,801	415,847	401,739	429,102	413,783	439,884
Cashflows at the Beginning of Period	0	399,270	639,033	906,541	1,159,999	1,440,801	1,856,648	2,258,387	2,687,489	3,101,272
Cashflows at the End of Period	399,270	639,033	906,541	1,159,999	1,440,801	1,856,648	2,258,387	2,687,489	3,101,272	3,541,155

Expected Balance Sheet

The balance sheet is one of the main statements which the project depends on, as it reflects the financial position of the entity and is usually estimated on the last day of the financial year.

All financial information and analysis of the project shall be the estimated for the years (2018-2027) as shown in the table below:

Table 5 Expected Balance Sheet

Description	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Assets										
Current Assets										
Cash	399,270	639,033	906,541	1,159,999	1,440,801	1,856,648	2,258,387	2,687,489	3,101,272	3,541,155
Inventory	0	0	0	0	0	0	0	0	0	0
Accounts Receivable	0	0	0	0	0	0	0	0	0	0
Total current Assets	399,270	639,033	906,541	1,159,999	1,440,801	1,856,648	2,258,387	2,687,489	3,101,272	3,541,155
Non Current Assets										
Fixed Assets (net)	1,839,910	1,742,084	1,643,945	1,545,480	1,446,677	1,347,523	1,248,005	1,148,108	1,047,820	947,123
Total Non Current Assets	1,839,910	1,742,084	1,643,945	1,545,480	1,446,677	1,347,523	1,248,005	1,148,108	1,047,820	947,123
Total Assets	2,239,179	2,381,117	2,550,486	2,705,479	2,887,478	3,204,171	3,506,391	3,835,597	4,149,091	4,488,278
Liabilities										
Current Liabilities										
Paybles	53,899	56,651	59,229	61,228	63,965	66,786	69,038	72,039	74,478	77,674
Remaining amount of Loan	75,920	83,118	90,997	99,624	(0)	(0)	(0)	(0)	(0)	0
Total current Liabilities	129,820	139,769	150,226	160,851	63,965	66,786	69,038	72,039	74,478	77,674
Non Current Liabilities										

Description	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Long Terms Loans	273,739	190,621	99,624	0	0	0	0	0	0	0
Total Long Term Liabilities	273,739	190,621	99,624	0	0	0	0	0	0	0
Total Liabilities	403,558	330,390	249,850	160,851	63,965	66,786	69,038	72,039	74,478	77,674
Owners Equity										
Shareholders Contributions	1,676,021	1,676,021	1,676,021	1,676,021	1,676,021	1,676,021	1,676,021	1,676,021	1,676,021	1,676,021
Statutory Reseve	15,960	37,471	62,461	86,861	114,749	146,136	176,133	208,754	239,859	273,458
Retained Profits	143,640	337,236	562,153	781,746	1,032,742	1,315,227	1,585,198	1,878,783	2,158,733	2,461,125
Total Equity	1,835,621	2,050,728	2,300,636	2,544,628	2,823,513	3,137,384	3,437,353	3,763,557	4,074,613	4,410,605
Total Liabilities and Equity	2,239,179	2,381,117	2,550,486	2,705,479	2,887,478	3,204,171	3,506,391	3,835,597	4,149,091	4,488,278

Feasibility Indicators

There are more than one method to calculate the discounted cash flow statement. However, in this study, WACC was used to calculate the discounted cash flows and net investment value. The Company's financing structure is based on equity and loans, therefore the discounted rate should be adjusted according to WACC based on the following assumptions, the following table illustrates the net expected free cash flow of the project:

Table 6 Free Net Cash flows Table

Net Cashflows	Pre- Operating	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027	Final value
Net Free Cashflows	(2,095,026)	362,071	348,831	376,576	362,526	389,869	415,847	401,739	429,102	413,783	439,884	4,230,901
Discount Factor	1.00	0.66	0.58	0.50	0.44	0.38	0.33	0.29	0.25	0.22	0.19	0.19
Net Present Value for Cash Flows	(2,095,026)	239,234	200,749	188,756	158,270	148,247	137,724	115,886	107,810	90,548	83,841	806,398

Table below (40) illustrates payback period for the project, as it is one of the indicators which investors care about before taking the investment decision as they need to know when the project will return the invested amount of money. Payback period definition is the total required period so that the project generates a total money equal to the total invested capital expenditure, as investor is always looking to return the full invested amount of money at the earliest possible.

Table 7 Payback Period

Payback Period	Pre- Operating	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Net Free Cashflows	(2,095,026)	362,071	348,831	376,576	362,526	389,869	415,847	401,739	429,102	413,783	439,884
Project Value Returned	2,095,026	1,732,955	1,384,124	1,007,548	645,021	255,152	(160,695)	(562,434)	(991,536)	(1,405,319)	(1,845,202)
Payback Period (Year)	5	1	1	1	1	1	0	0	0	0	0

Table :8 Financial Analysis Results

Indicator	Value
Weighted Average Cost of Capital (WACC)	14.81%
Net Present Value for Cashflows	182,437
Payback Period	5
Internal Rate of Return	20.66%

Sensitivity Analysis

Sensitivity analysis is conducted to investigate the effects of changes in significant inputs of the financial analysis. This includes changes in revenues, total investment costs, and operating costs.

Table 9 Sensitivity Analysis

Sensitivity Analysis	Internal Rate of Return	Payback Period	WACC
Original Scenario	%20.66	5	%14.81
Revenues declined by 10%	%16.25	6	%14.81
Operating Expenses Increased by 10%	%19.01	6	%14.81

Conclusions and Recommendations

Based on the profitability analysis and considering that project internal rate of return is higher than WACC, it is concluded that the project is feasible as the net present value for the project is positive 182,437 Jordan Dinars considering that the project provides 66 Job Opportunities for the governorate residents.